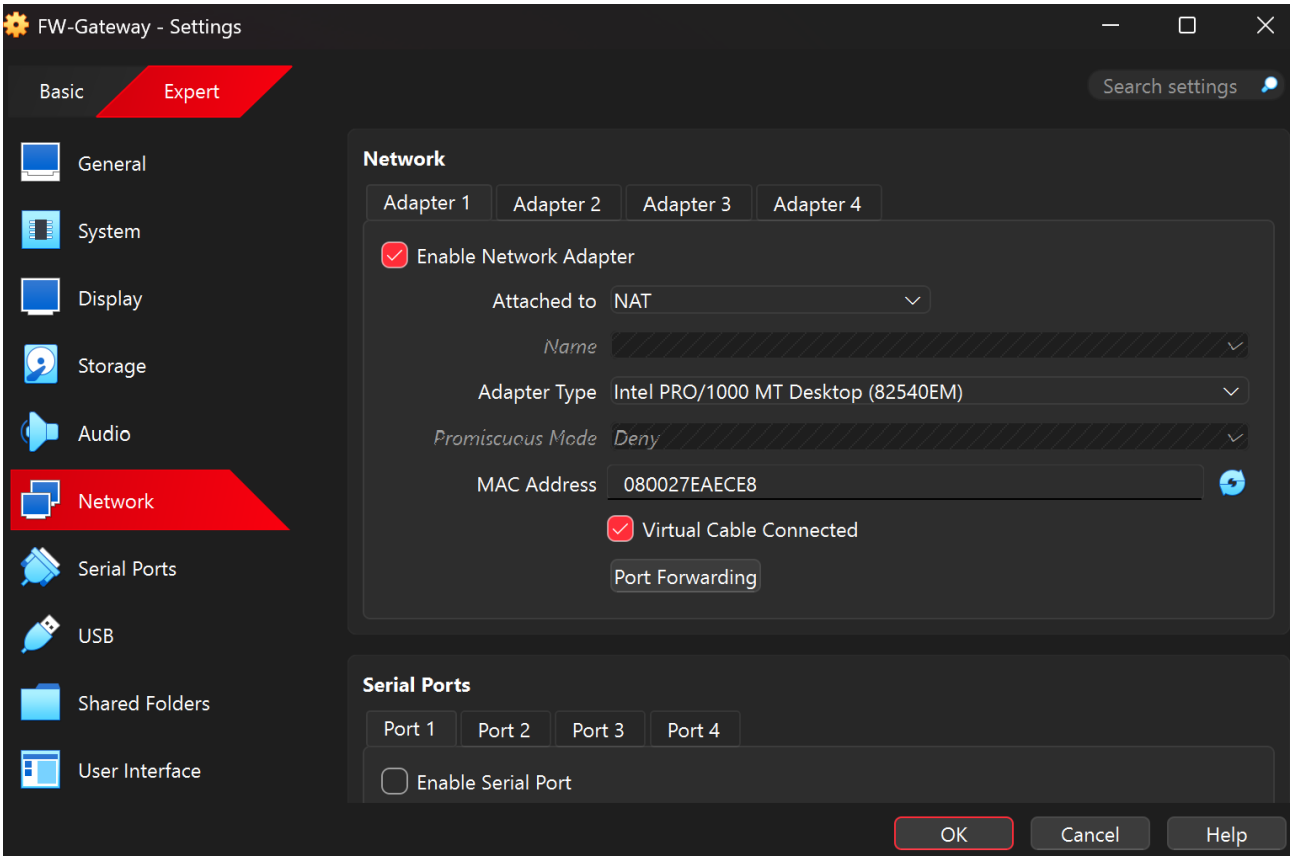
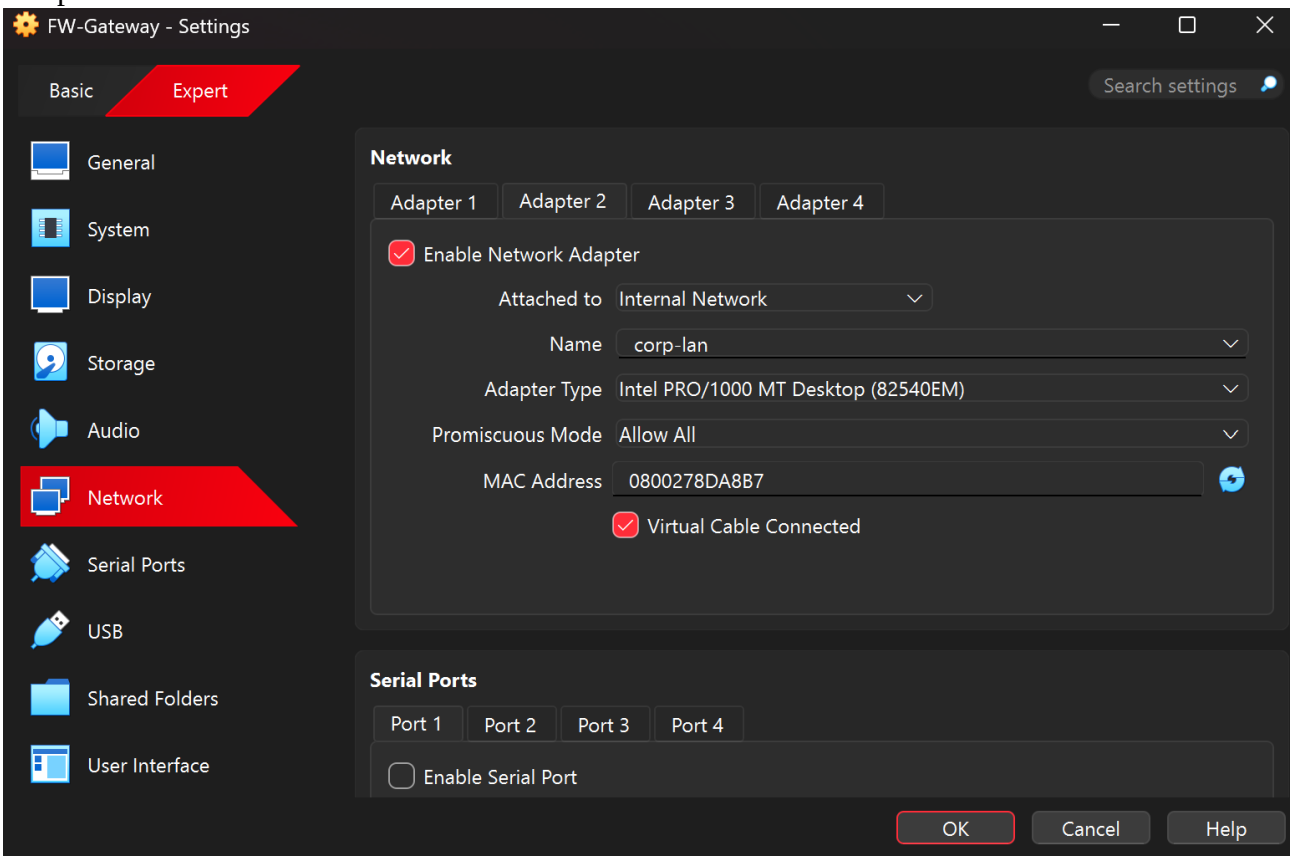
FW-Gateway Network Configurations:

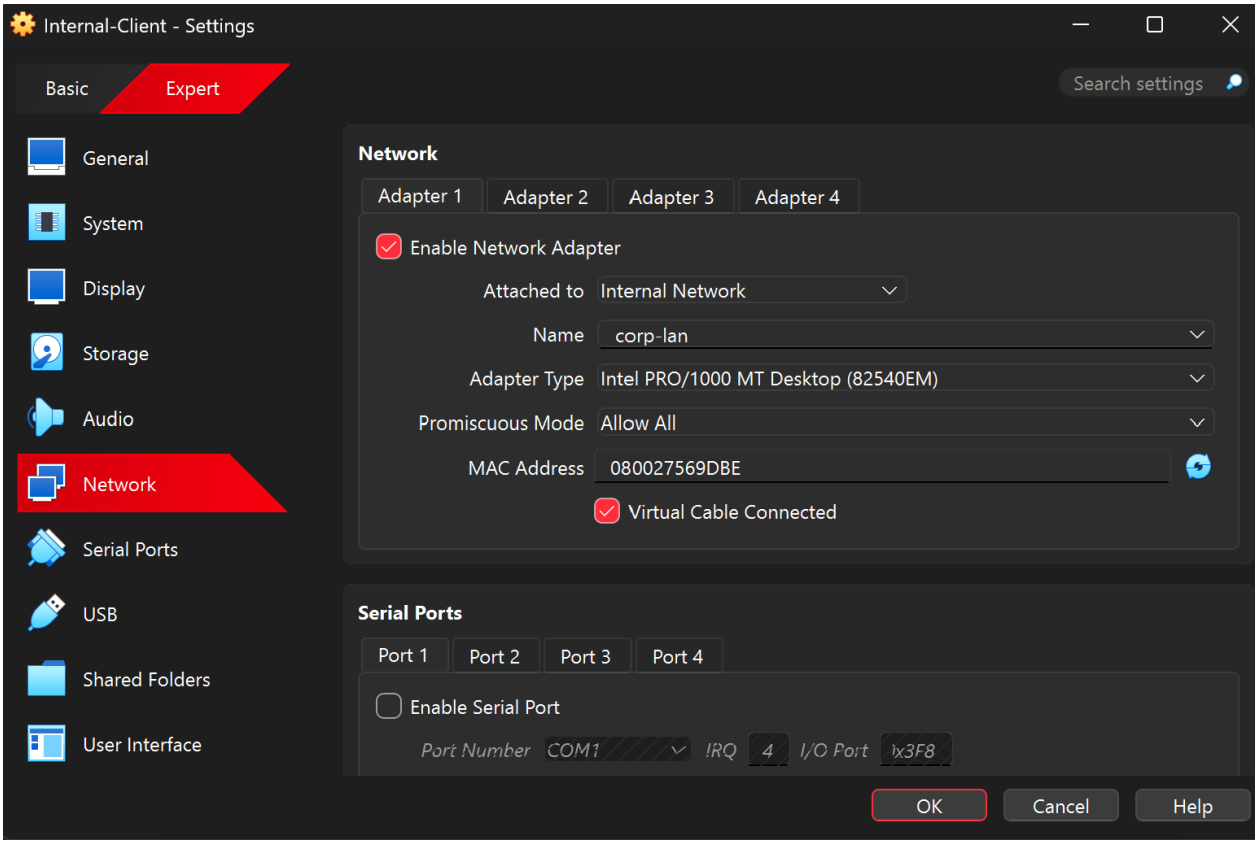
Adapter 1:



Adapter 2:

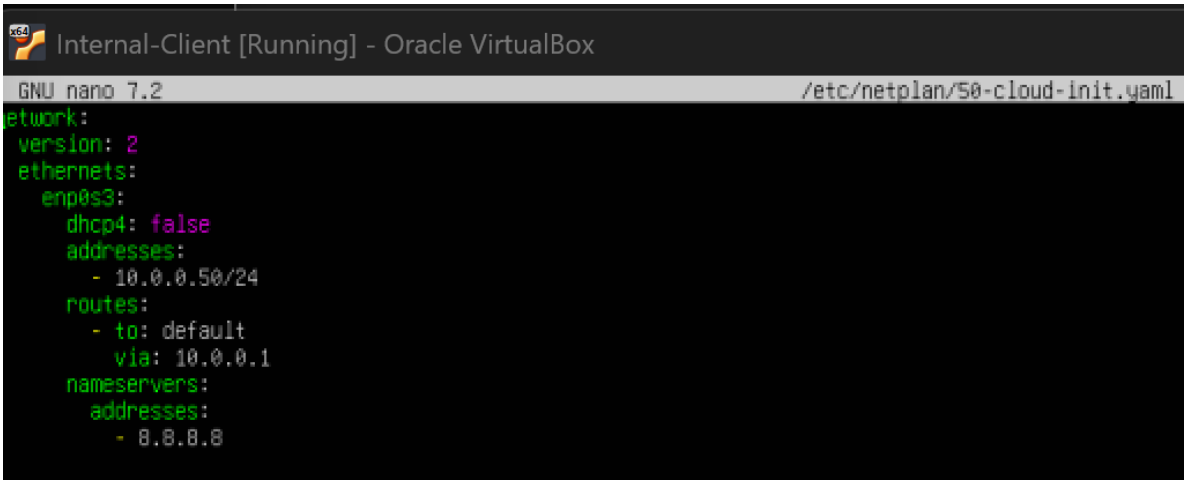


Internal-Client Network Configurations:

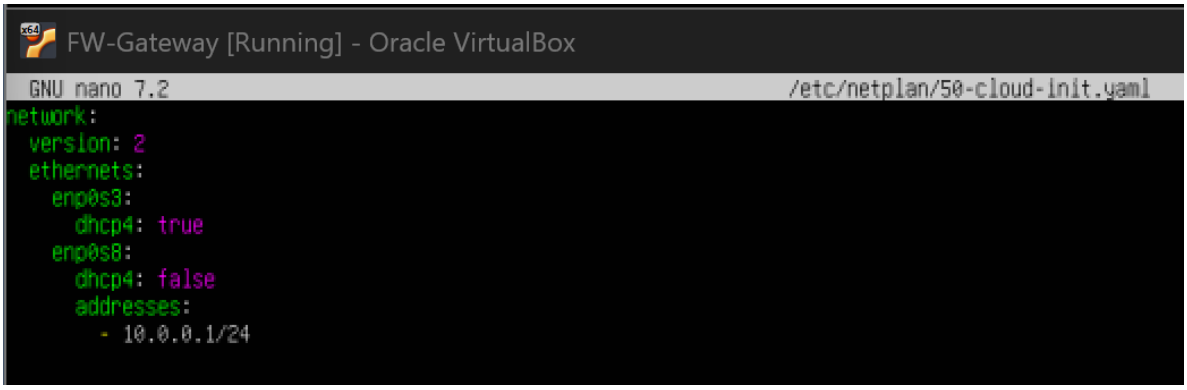


Demonstrates the hypervisor interface mapping where FW-Gateway is configured with a public NAT adapter (WAN) and a private Internal Network adapter named corp-lan. The Internal-Client is tied exclusively to corp-lan, isolating it from direct internet access.

Netplan Configurations on Internal-Client:



Netplan Configurations on FW-Gateway:



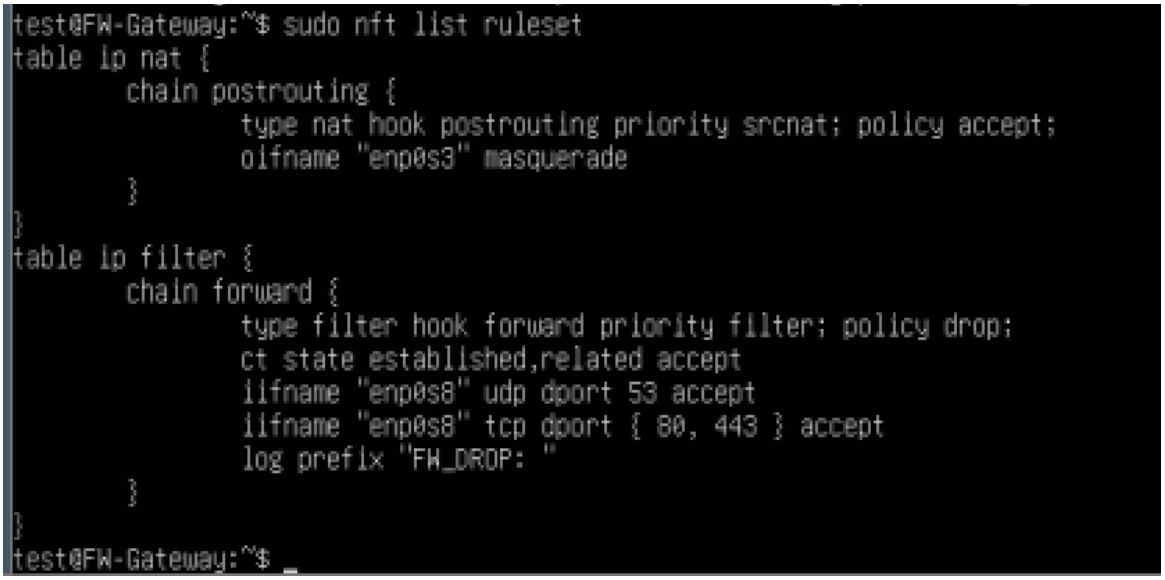
Shows the static network. The gateway binds 10.0.0.1/24 to its LAN interface (enp0s8), and the client maps 10.0.0.50/24 to enp0s3 with its default route targeting the gateway.

IP Forwarding:



Displays the successful reload of system control parameters forcing net.ipv4.ip_forward = 1. This safely transitions the Linux kernel from an endpoint architecture into an active Layer 3 transit router.

NFTables Configuration:



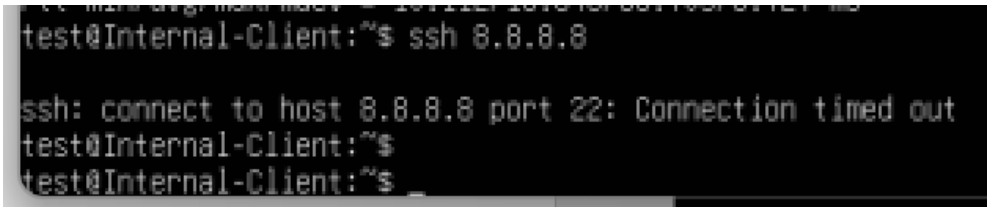
Documents the loaded runtime firewall infrastructure inside the kernel. It highlights the independent NAT masquerade hook paired with a stateful filter forward chain executing an explicit default-DROP policy.

Client Connectivity (PASS):

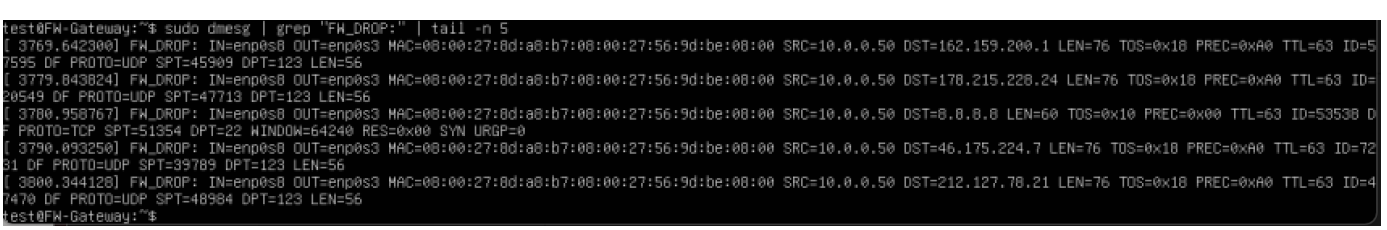


Validates the permitted egress configuration paths from the private network segment. The structural HTTP header return proves that outbound DNS (UDP 53) and HTTPS web traffic (TCP 443) are explicitly allowed and functioning.

Client connectivity SSH Fails:



Matching FW-DROP log entry:



Captures the firewall dropping an unauthorized outbound SSH connection attempt (TCP 22) from the client machine. The gateway kernel immediately catches and logs the violation using the custom FW_DROP: alert string.

- **What Netfilter is and how it resides within the Linux kernel:** Netfilter is a built-in security tool inside the Linux operating system kernel. It works by placing five checkpoints along the network path to check data packets exactly as they travel through the system.

- **The difference between the Netfilter framework and the nftables command-line utility**: Netfilter is the invisible engine running inside the system that tracks connections and decides whether to block or allow data. The nftables command is simply the menu or tool that a human uses to type in the rules and send them to that engine.

- **The architecture of Tables, Chains, and Rules:** The system is organized like a set of folders. Tables act as main folders for different types of traffic. Inside tables are Chains, which act as subfolders tied to specific checkpoints with instructions on what to do by default, such as drop everything. Inside chains are individual Rules that look for specific triggers, like a port or network interface, and give a final verdict to accept, drop, or log the data.

- **Why modern Linux relies on this rather than legacy tools like TCP Wrappers (hosts.allow / hosts.deny):** Old tools like TCP Wrappers were slow because they worked high up at the application level and only protected specific programs that were built to support them. Modern Netfilter runs directly inside the system memory at the network level, allowing it to protect all types of traffic instantly with much faster performance.